

12. The following data refer to two planets.

	radius / km	density / kg m ⁻³
Planet P	2000	6000
Planet Q	16000	3000

The gravitational field strength at the surface of P is 13.4 N kg⁻¹.

What is the gravitational field strength at the surface of Q?

- A** 3.4 N kg⁻¹ **B** 13.4 N kg⁻¹ **C** 53.6 N kg⁻¹ **D** 80.4 N kg⁻¹

Answer: C

$$\text{gravitational field strength } g = \frac{GM}{R^2} = \frac{G}{R^2} \left(\frac{4}{3} \pi R^3 \rho \right) = \frac{4\pi G \rho}{3} R \Rightarrow g \propto \rho R$$

$$\text{so, } \frac{g_Q}{g_P} = \frac{\rho_Q}{\rho_P} \times \frac{R_Q}{R_P} = \frac{3000}{6000} \times \frac{16000}{2000} = 4 \quad g_Q = 4 \times 13.4 = 53.6 \text{ N kg}^{-1}$$